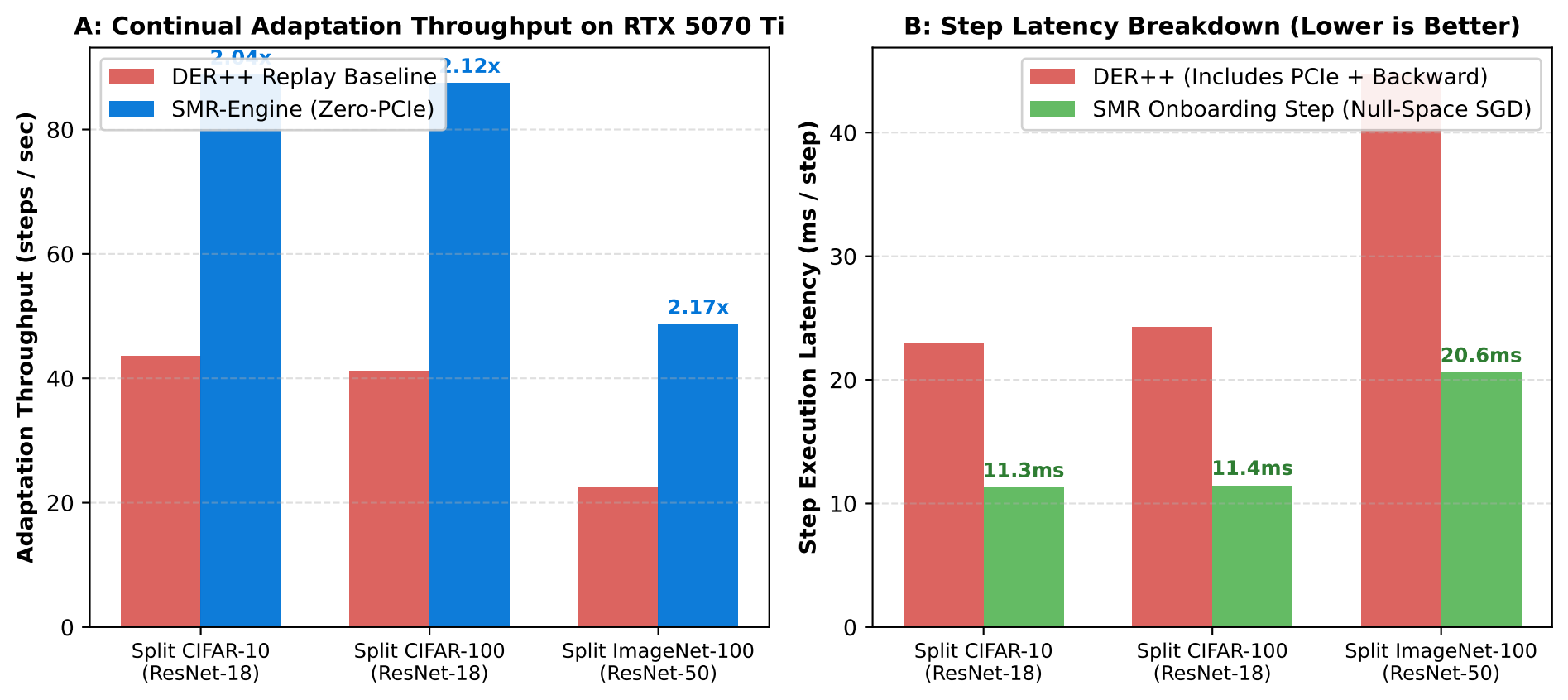
A: Continual Adaptation Throughput on RTX 5070 Ti
DER++ Replay Baseline
SMR-Engine (Zero-PCIe)
Adaptation Throughput (steps / sec)
0
20
40
60
80
2.04x
2.12x
2.17x
Split CIFAR-10
(ResNet-18)
Split CIFAR-100
(ResNet-18)
Split ImageNet-100
(ResNet-50)
B: Step Latency Breakdown (Lower is Better)
DER++ (Includes PCIe + Backward)
SMR Onboarding Step (Null-Space SGD)
Step Execution Latency (ms / step)
0
10
20
30
40
11.3ms
11.4ms
20.6ms
Split CIFAR-10
(ResNet-18)
Split CIFAR-100
(ResNet-18)
Split ImageNet-100
(ResNet-50)